

AI-Assisted Regulatory Intelligence and Adaptive Compliance Verification System for Banking Supervision

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Abstract—Regulatory requirements issued by the Reserve Bank of India (RBI) span numerical thresholds, deterministic conditions, temporal obligations, and qualitative standards such as “adequate due diligence,” each of which demands a different mode of verification. Bank employees and compliance officers must first locate and interpret the applicable requirement within a large and continually growing corpus of Master Directions, circulars, and notifications, while the regulator must independently verify, at scale, whether submitted bank evidence satisfies that requirement. We present a two-module regulatory intelligence system that addresses both sides of this problem within a single pipeline. The Regulation Comprehension module ingests an RBI circular or Master Direction, segments and classifies its clauses, extracts a typed and machine-readable rule representation, simplifies regulatory language for human readers, and supports natural-language question answering with clause-level source citation. The Adaptive Compliance Checking module reuses this same structured regulatory knowledge to evaluate bank-submitted evidence against applicable rules from the supervisory perspective of RBI, returning one of five decisions – COMPLIANT, NON_COMPLIANT, INSUFFICIENT_DATA, AMBIGUOUS, or REQUIRES_HUMAN_REVIEW – together with an evidence trace. The central technical contribution is an adaptive reasoning-modality router that decides, per regulatory requirement and per evidence set, whether a deterministic rule engine, a retrieval-grounded large language model (LLM), or human escalation should produce the decision, governed by an explicit cost–accuracy–latency objective and a hard architectural override that sends fully specified deterministic rules with complete evidence to the rule engine unconditionally. This differs from the well-studied problem of routing between LLMs of different capability, since our router selects among fundamentally different reasoning mechanisms rather than among models. We further contribute a rule-grounded synthetic RBI compliance benchmark spanning nine case-difficulty categories and a separate, dedicated regulation-comprehension benchmark (extraction, question-answering, and simplification sub-benchmarks), so that the two modules are evaluated with equal rigor rather than treating comprehension as a subordinate preprocessing step; this

yields two independent sets of metrics, baselines, and ablations, one per module, spanning extraction accuracy, retrieval and answer-faithfulness metrics, simplification quality, compliance accuracy, and routing efficiency. We further contribute an evidence-grounded explanation architecture that traces every decision and every comprehension answer to a specific rule, clause, and evidentiary or source field. Because operational bank data cannot be obtained for confidentiality and regulatory reasons, all quantitative figures reported in this draft’s results section are explicitly marked as illustrative placeholders rather than experimental findings, and are intended to be replaced once the pipelines described in this paper are implemented and evaluated.

Index Terms—RegTech, RBI regulations, regulatory intelligence, regulatory natural language processing, automated compliance checking, neuro-symbolic AI, retrieval-augmented generation, large language models, adaptive reasoning, banking supervision

I. INTRODUCTION

Banking regulators such as the Reserve Bank of India publish and continually revise a large corpus of Master Directions, circulars, notifications, and frequently-asked-question clarifications that together define compliant conduct for scheduled commercial banks and non-banking financial companies. Two distinct groups of readers must engage with this corpus, and each faces a different obstacle. Bank employees, compliance officers, and internal auditors must locate the specific requirement relevant to a given operational question, interpret dense legal and regulatory language correctly, and reconcile amendments and supersessions across multiple related documents – a task that is slow, error-prone, and does not scale with the pace at which the regulations themselves are revised. The regulator, in turn, must supervise a large number of institutions, each periodically submitting audit reports, compliance filings, and

operational records, and must determine whether the submitted evidence in fact satisfies the applicable regulatory requirement – a task that resists uniform automation because the requirements themselves are heterogeneous.

That heterogeneity is the paper’s starting observation. RBI regulatory text mixes numeric thresholds and ratios (for example, exposure limits expressed as a percentage of eligible capital), Boolean and procedural conditions (for example, the existence of a board resolution), temporal obligations (for example, a report that must be filed within a stated number of days), and qualitative standards phrased in natural language (for example, “reasonable measures” or “adequate due diligence”) whose satisfaction cannot be reduced to a closed-form numeric test. A system that treats all of these uniformly misallocates reasoning effort in one of two directions. Delegating every check to a large language model inherits hallucination risk precisely on the numeric, high-stakes checks where a deterministic answer is both achievable and necessary. Conversely, restricting the system to hand-authored deterministic rules leaves the qualitative and context-dependent share of RBI regulatory text – which is substantial – entirely unaddressed.

This paper is organized around two integrated modules that respond to the two reader groups described above. The Regulation Comprehension module addresses the human-facing problem: it extracts, structures, simplifies, and makes searchable the content of an RBI circular so that a bank employee or auditor can ask a natural-language question and receive an answer grounded in a specific clause and source document. The Adaptive Compliance Checking module addresses the supervisory problem: it reuses the same structured regulatory knowledge to evaluate bank-submitted evidence, and it is here that the paper’s principal technical question arises – given a regulatory requirement and the evidence available about a bank, which reasoning mechanism should answer it, and how should that decision be made in a principled, auditable way rather than through an arbitrary confidence threshold. We refer to this problem as reasoning-modality routing, to distinguish it explicitly from the substantial existing literature on routing between large language models of differing capability and cost, which selects among models for a fixed reasoning strategy. Our routing problem instead selects among qualitatively different reasoning strategies – a deterministic rule engine, a retrieval-grounded LLM, and human escalation – for a fixed underlying model.

A. Contributions

This paper makes the following contributions, ordered by the primacy assigned to them in Section XVIII.

- An adaptive, cost-aware reasoning-modality router for regulatory compliance checking, formalized as a constrained-optimization problem over rule-engine, retrieval-augmented-LLM, and human-review paths, evaluated against static, confidence-threshold, and fixed-hybrid routing baselines on an accuracy–cost–latency frontier (primary contribution).
- A schema-constrained rule-extraction pipeline, situated within a broader Regulation Comprehension module, that converts RBI Master Direction and circular clauses into a typed, composable, executable rule representation covering quantitative, Boolean, conditional, temporal, qualitative, relational, and multi-entity requirement categories, with per-category extraction accuracy to be reported once evaluated (secondary contribution).
- A systematically constructed, rule-grounded synthetic compliance benchmark covering nine case-difficulty categories – compliant, non-compliant, boundary, near-boundary, ambiguous, missing-data, adversarial, contradictory, and cross-version – addressing the absence of a public RBI-specific compliance benchmark (secondary contribution).
- A dedicated regulation-comprehension evaluation methodology – extraction, question-answering, and simplification sub-benchmarks, each with its own baselines and metrics (Sections XIII, XIV.B, XV-B) – so that Module 1 is measured with the same rigor as Module 2 rather than assumed to work because Module 2 works (secondary contribution).
- An evidence-grounded decision architecture in which every compliance decision, and every answer returned by the regulation comprehension question-answering interface, is traceable to a specific rule identifier, regulatory clause, and evidentiary or source input (supporting contribution).
- A system-level integration in which the structured regulatory knowledge produced by Regulation Comprehension is directly reused by Adaptive Compliance Checking, forming a single regulatory intelligence pipeline rather than two disconnected applications (system-level contribution).

B. What this paper does not claim

Consistent with the novelty analysis in Section XVIII, this paper does not claim that combining a rule engine with retrieval-augmented LLM reasoning is itself novel; comparable hybrid architectures exist in recent regulatory-compliance natural language processing work (Section III). It does not claim that rule extraction from regulatory text is a new technique; that technique has a well-established lineage in the automated compliance checking literature for construction codes, applied here to a new domain. It does not claim that retrieval-augmented generation, large language model use, or generic routing between language models are novel in themselves. It does not implement or evaluate connected-counterparty graph reasoning for exposure aggregation; Section XX discusses this as a theoretical extension only, because a meaningful evaluation would require multi-hop ownership data that is not publicly available. It does not implement optical character recognition or scanned-document ingestion; bank-evidence inputs are restricted to structured formats and PDFs with extractable text.

C. Paper organization

Section II formalizes the two-part research problem and states the research questions. Section III reviews related work and identifies the specific gap addressed. Section IV presents the integrated system architecture. Section V describes the Regulation Comprehension module in full technical detail. Section VI details the regulatory rule representation. Section VII describes the Adaptive Compliance Checking module in comparable detail. Section VIII presents the adaptive reasoning-modality router. Section IX discusses evidence-grounded explainability. Section X discusses the end-to-end integration of both modules. Section XI describes implementation. Section XII presents the experimental methodology for Adaptive Compliance Checking, and Section XIII the parallel methodology for Regulation Comprehension. Section XIV presents baselines for both modules, and Section XV (with Section XV-B) the evaluation metrics for both. Section XVI reports illustrative placeholder results for both modules. Section XVII discusses the planned ablation studies for both modules. Section XVIII assesses novelty. Section XIX discusses limitations and threats to validity, Section XX future work, and Section XXI concludes.

II. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

A. Regulation comprehension

Let D denote an RBI regulatory document (a Master Direction, circular, or notification). The regulation comprehension task is to produce a structured representation of D – clause segments, definitions, applicability conditions, temporal obligations, exceptions, and category labels – together with a simplified natural-language rendering suitable for a non-legal reader, and to support natural-language question answering over D such that every answer cites the specific clause and source document that supports it.

B. Regulatory compliance verification

Let $R = \{r_1, \dots, r_n\}$ denote the set of regulatory rules extracted from an RBI regulatory corpus (Section VI), where each rule r_i is represented as the typed object defined in Section VI. Let E denote a bank entity’s evidence set: the structured and unstructured data available about that entity relevant to a given rule. A compliance query is a pair (r_i, E) . The compliance-checking task is to produce a decision

$d \in \{\text{COMPLIANT}, \text{NON_COMPLIANT}, \text{INSUFFICIENT_DATA},$

$\text{AMBIGUOUS}, \text{REQUIRES_HUMAN_REVIEW}\},$

(2)

together with an evidence trace τ identifying the specific rule identifier, clause reference, and evidence fields that justify d .

C. Research objectives and research questions

The overall objective is to determine whether a single reasoning mechanism should never be applied uniformly across RBI regulatory requirements, and whether an adaptively

routed combination of a deterministic rule engine, a retrieval-grounded LLM, and human escalation can achieve higher compliance accuracy at comparable or lower cost and latency than any single mechanism used alone. We pose the following research questions.

- **RQ1 (Compliance accuracy):** Does adaptive reasoning-modality routing improve overall and per-case-type compliance accuracy relative to rule-only, LLM-only, retrieval-augmented-LLM-only, and fixed-hybrid baselines?
- **RQ2 (Routing efficiency):** Does the proposed router achieve a more favorable accuracy–cost–latency trade-off than static, confidence-threshold, and fixed-hybrid routing strategies?
- **RQ3 (Rule and regulation extraction quality):** What is the precision, recall, and executability of rules and structured regulatory knowledge extracted from RBI text, broken down by requirement category?
- **RQ4 (Evidence grounding and hallucination):** How much does evidence grounding – in both the regulation comprehension question-answering interface and the compliance decision architecture – reduce hallucinated or unsupported outputs relative to an ungrounded LLM baseline, particularly on adversarial and missing-data cases?
- **RQ5 (Robustness to difficult cases):** How does system behavior degrade, if at all, across boundary, near-boundary, ambiguous, missing-data, adversarial, contradictory, and cross-version cases relative to compliant and non-compliant cases?
- **RQ6 (Regulation comprehension quality):** How accurately does the system retrieve the clause(s) relevant to a natural-language regulatory question, and how faithful and relevant are the generated answers, relative to keyword-search, LLM-only, and generic-RAG baselines?
- **RQ7 (Simplification fidelity):** Does RBI-specific clause simplification achieve a meaningful readability improvement over the source clause without a corresponding loss in the fidelity of stated conditions and exceptions?

III. RELATED WORK

A. Automated compliance checking and rule extraction from regulatory text

Automated compliance checking has a substantial research history predating large language models, concentrated in the construction and building-code domain. Zhang and El-Gohary [1] propose a semantic, pattern-based natural language processing approach for extracting information-extraction and conflict-resolution rules from construction regulatory documents, reducing the number of hand-authored extraction patterns required through syntactic and semantic text features. A related line of work applies ontology- and rule-based natural language processing to interpret utility-infrastructure regulations, decomposing text into requirement, applicability, selection, and exception components [2]. More recently, Hettiarachchi et al. [3] introduce a manually annotated corpus of

building-regulation sentences with entity and relation labels intended to support machine-readable rule generation with modern natural language processing and large language model methods, motivating a shift from hand-authored extraction patterns toward learned extraction as regulatory domains multiply. This body of work establishes the feasibility and the standard evaluation methodology – entity, operator, and threshold precision/recall/F1, together with rule executability – for extracting structured rules from regulatory natural language, a methodology we adopt and adapt to RBI banking regulation in Section V without claiming to have originated it.

B. Neuro-symbolic and hybrid regulatory compliance systems

A parallel and more recent line of work applies hybrid neural and symbolic architectures directly to financial and regulatory compliance. Ershov [4] demonstrates the construction of an executable regulatory knowledge graph from a BERT-tagged corpus of Abu Dhabi Global Market regulations, using coreference resolution and syntactic analysis to populate a graph database queried through a graph query language, and proposes graph neural networks as a future direction for judgment automation. Subsequent work explicitly frames compliance reasoning as knowledge-graph-augmented retrieval-augmented generation, monitoring regulatory change, extracting structured obligations, and detecting compliance gaps against internal policy jointly across multiple regulatory frameworks, and reports grounding-accuracy figures as a way of quantifying hallucination mitigation from knowledge-graph grounding [5]. Separately, symbolic reasoning layers over LLM-constructed regulatory knowledge graphs have been proposed for anti-money-laundering-style compliance obligations [6], and graph-neural-network-based transaction monitoring combined with an LLM explanation layer has been evaluated on simulated transaction streams for compliance flagging [7]. Code generation as an intermediate, executable representation for financial compliance checking has also been explored [8]. Collectively, this literature establishes that grounding LLM compliance reasoning in an extracted symbolic representation is an active, multiply-instantiated research direction, which materially narrows the novelty available to a system that merely combines these components (Section XVIII).

C. Regulatory question answering and regulation comprehension

Beyond compliance decisioning, a related strand of work targets the human-facing problem of understanding regulatory text itself: converting dense legal or regulatory language into a form a non-specialist reader can query and interpret. Retrieval-augmented generation, in which a generator is conditioned on passages retrieved from an external corpus rather than relying solely on parametric knowledge, was introduced by Lewis et al. [24] as a general-purpose mechanism for grounding generation in retrievable evidence, and is the dominant architectural pattern for regulatory and legal question answering in the

broader literature (Section III-G). Feeding this pipeline requires first recovering a document’s textual and structural content faithfully; layout-aware document understanding models such as LayoutLM [23], which jointly model text and layout signals rather than treating a document as a flat token stream, establish that structural cues (headings, paragraph numbering, tabular layout) materially improve downstream extraction from regulatory-style documents, motivating the layout-aware ingestion step of Regulation Comprehension (Section V). Because a retrieval-augmented generator is not guaranteed to stay faithful to what was retrieved, a further strand of work addresses evaluating and mitigating this gap: reference-free RAG evaluation frameworks such as RAGAs [25] decompose RAG quality into retrieval-side metrics (context precision, context recall) and generation-side metrics (faithfulness, answer relevance), which we adopt as the basis for the comprehension-side evaluation methodology in Section XIII; and the broader survey of hallucination in natural language generation by Ji et al. [26] catalogs the mitigation strategies – of which citation-constrained generation and retrieval grounding are two – that motivate the hallucination-mitigation design of Section V. The Regulation Comprehension module in this paper follows this retrieval-plus-citation pattern applied specifically to RBI Master Directions and circulars, including their category structure (KYC, anti-money-laundering, lending, transaction reporting, governance, audit), amendment relationships, and effective-date semantics, which we do not find addressed jointly in a single published system for this jurisdiction. Legal contract-understanding datasets such as CUAD [28], which pose clause identification and extraction as a span-based question-answering task over more than 500 expert-annotated commercial contracts, and financial numerical-reasoning benchmarks such as FinQA [29], which require multi-step arithmetic reasoning grounded in tabular and textual financial-report content, establish the general evaluation pattern – expert-annotated ground truth, span- or program-level correctness, and citation to source text – that Section XIII adapts for RBI regulatory comprehension, though neither dataset targets banking-regulatory text specifically.

D. RBI-specific and Indian banking compliance automation

Commercial governance-risk-compliance platforms marketed for RBI compliance automation, including regulatory-mapping and reporting-automation products, are widely available but describe product capability rather than a reproducible research methodology, benchmark, or evaluation [9]. We were unable to locate a peer-reviewed system that targets RBI regulatory text specifically, evaluates rule- or regulation-extraction accuracy against RBI’s clause structure across Master Directions, circulars, and FAQs including amendment and supersession relationships, or publishes a benchmark for RBI compliance reasoning. We treat this as the primary domain-application gap addressed by this paper, while remaining explicit that the underlying techniques (Sections III.A–III.B) are not themselves new.

E. Retrieval-augmented generation and LLM-based regulatory reasoning

Retrieval-augmented generation, in which a language model’s output is conditioned on passages retrieved from an external corpus rather than generated purely from parametric knowledge, is now the standard mechanism for grounding LLM outputs in source text across legal, financial, and regulatory domains, and underlies both the question-answering component of Regulation Comprehension (Section V) and the qualitative-reasoning path of Adaptive Compliance Checking (Section VII). We do not claim retrieval-augmented generation itself as a contribution; our use of it is standard, and the paper’s contribution lies instead in when the system chooses to invoke it relative to deterministic reasoning (Section VIII).

F. LLM routing

A large and fast-moving literature addresses routing a query to the most cost-effective sufficiently capable large language model from a pool of candidates. Hybrid LLM [10] and BEST-Route [11] formalize routing under quality-cost constraints; RouteLLM [12] learns routing policies from human preference data; GraphRouter [13] and universal model routing [14] represent models and queries jointly for learned selection; cascade- and uncertainty-based routers query a cheap model first and escalate only on low confidence [15]; and survey and benchmarking work provides standardized evaluation protocols for this line of work [16]. This literature is, almost without exception, concerned with selecting among LLMs of varying size and cost for a single reasoning strategy. Our routing problem (Section VIII) is structurally different: it routes among qualitatively distinct reasoning modalities – a deterministic symbolic engine, a retrieval-grounded LLM, and a human escalation path – for a fixed underlying model, motivated not by inference cost alone but by which modality can answer a given requirement with a defensible correctness guarantee. We adopt this literature’s evaluation standard, reporting an accuracy–cost–latency frontier against static, threshold, and learned baselines rather than a single aggregate accuracy figure (Sections XII–XVI), while arguing that the problem itself has not been addressed in this form for compliance reasoning.

G. Synthetic benchmarks for financial and regulatory AI

Recent work argues that static question-answer benchmarks are insufficient validation for financial and compliance-oriented LLM systems, and calls explicitly for benchmarks incorporating retrieval evidence requirements, escalation scenarios, and adversarial cases [17]. Logic-guided synthesis has been used to construct test cases that probe implicit regulatory-compliance violations in LLM tool invocation across financial, healthcare, and Internet-of-Things domains [18], and semi-synthetic benchmark generation has been proposed specifically to produce contamination-resistant financial evaluation data [19]. Our benchmark construction methodology (Section XII) follows this emerging practice – generating test cases systematically from extracted rules rather than independently, and explicitly including boundary, adversarial, missing-data,

and cross-version cases – while, to our knowledge, being among the first such benchmarks grounded specifically in RBI regulatory text. Our extraction and retrieval components also build on general legal-natural-language-processing foundations, including domain-adapted language models [20] and benchmark suites for legal reasoning and legal language understanding [21], [22], which establish that general-purpose language understanding transfers imperfectly to legal and regulatory text.

H. Research gap

No existing system, to our knowledge: (a) formalizes and evaluates routing between deterministic and neural reasoning modalities, as opposed to routing between neural models, for compliance checking; (b) applies rule extraction and hybrid reasoning specifically to RBI regulatory text with a reported per-category accuracy breakdown, jointly with a human-facing regulation comprehension interface built from the same extracted representation; or (c) provides a public, rule-grounded synthetic benchmark for Indian banking regulatory compliance. This paper targets all three, with (a) as its primary claim. Table I summarizes how the proposed system relates to the categories of prior work discussed above.

IV. SYSTEM OVERVIEW

The system consists of two interconnected modules rather than two independent applications. Regulation Comprehension (Section V) converts an RBI regulatory document into structured, machine-readable regulatory knowledge and a human-facing question-answering interface. Adaptive Compliance Checking (Section VII) consumes that same structured knowledge to evaluate bank-submitted evidence from RBI’s supervisory perspective. The two modules share a single regulatory knowledge base rather than maintaining separate representations, which is the basis of the integration contribution discussed in Section X.

The complete pipeline, illustrated in Fig. 1, proceeds as follows: an RBI circular or Master Direction is ingested and processed; Regulation Comprehension extracts structured regulatory knowledge from it; that knowledge simultaneously populates a regulatory search and question-answering interface for human users and a compliance rule repository consumed by Adaptive Compliance Checking; bank evidence is separately ingested and normalized; the Adaptive Compliance Router determines, per rule and per evidence set, whether the deterministic rule engine, the retrieval-augmented LLM, or human review should produce the compliance decision; evidence is aggregated and a decision is assigned; and an audit-ready supervisory report is generated, citing the applicable rule, clause, evidence, reasoning modality used, and any recommended corrective action. Table II summarizes the system’s processing layers and the role of each.

A. Scope decisions

Two scope decisions materially affect what this paper implements and evaluates, and are stated explicitly rather than

TABLE I
COMPARISON WITH CATEGORIES OF EXISTING APPROACHES DISCUSSED IN SECTION III.

Approach	Regulation Understanding	Rule-Based Verification	LLM Reasoning	Adaptive Modality Routing	RBI-Specific	Human Escalation	Evidence Traceability
Construction-code ACC [1]–[3]	Partial	Yes	No	No	No	No	Partial
Knowledge-graph RAG compliance [4]–[7]	Yes	Partial	Yes	No	No	No	Yes
Compliance-to-Code [8]	Partial	Yes	Yes	No	No	No	Partial
Commercial RBI GRC platforms [9]	Yes	Yes	Unknown	Unknown	Yes	Unknown	Unknown
LLM-to-LLM routing [10]–[16]	No	No	Yes	No (routes models, not modalities)	No	No	No
Layout-aware document QA [23], [24], [28], [29]	Yes	No	Yes	No	No	No	Partial
Proposed system	Yes	Yes	Yes	Yes	Yes	Yes	Yes

TABLE II
SYSTEM PROCESSING LAYERS.

Layer	Function	Module	Processing stage
1	Regulatory document ingestion (PDF/HTML acquisition, cleaning, section identification)	Regulation Comprehension	Offline
2	Clause segmentation, definition extraction, cross-reference and effective-date detection	Regulation Comprehension	Offline
3	Rule extraction and classification into the typed schema (Section VI)	Regulation Comprehension	Offline
4a	Relational rule store: structured rules, versions, applicability, thresholds	Shared knowledge base	Offline
4b	Vector store: clause and definition embeddings for semantic retrieval	Shared knowledge base	Offline
5	Regulatory search / question answering with clause-level citation	Regulation Comprehension	Query-time
6	Bank evidence ingestion and canonicalization into structured evidence fields	Adaptive Compliance Checking	Query-time
7	Adaptive reasoning-modality router (Section VIII)	Adaptive Compliance Checking	Query-time
8	Reasoning: deterministic rule execution or retrieval-grounded LLM reasoning	Adaptive Compliance Checking	Query-time
9	Evidence aggregation, conflict detection, and compliance decision assignment	Adaptive Compliance Checking	Query-time
10	Explainability: audit-ready supervisory report generation	Adaptive Compliance Checking	Query-time

system_architecture.png

Fig. 1. End-to-end architecture of the proposed regulatory intelligence and compliance verification system.

left implicit. First, connected-counterparty graph reasoning for large-exposure aggregation across related entities is not implemented; Section XX discusses it as a theoretical extension, because its evaluation would require multi-hop ownership data that is not publicly available, which would make any accuracy claim dependent on synthetic ownership data generated by the

authors themselves – a validation loop judged too weak to report as a result. Second, optical character recognition and scanned-document ingestion are out of scope for the evaluated system; both regulatory-document and bank-evidence inputs are restricted to formats with extractable text (structured CSV/JSON and text-layer PDFs), since OCR quality is an orthogonal engineering concern that does not bear on the research questions of Section II.

V. REGULATION COMPREHENSION MODULE

This section describes the module that converts an RBI regulatory document into structured knowledge and a human-facing comprehension interface, and whose output is reused directly by Adaptive Compliance Checking (Section X). We describe this module in the same technical depth as Adaptive Compliance Checking (Section VII) and evaluate it with a dedicated methodology, metrics, and baselines (Section XIII), rather than treating it as a preprocessing step subordinate to compliance checking. The two modules are measured independently because they answer different questions – “what does the regulation say and mean” versus “does this bank satisfy it” – and a system that comprehends regulation well but checks compliance poorly, or vice versa, is a materially different (and differently deployable) system from one that does both well; collapsing the two into a single reported number would hide that distinction.

A. RBI document ingestion

The module accepts an RBI circular or Master Direction supplied as a PDF, together with associated metadata (document title, issue date, document type – Master Direction,

Master Circular, or standalone circular – and, where available, the department of RBI that issued it, such as the Department of Regulation or the Department of Banking Supervision). Ingestion is idempotent: re-submitting an already-ingested document is detected by content hash and does not create a duplicate entry in the knowledge base, which matters operationally because RBI frequently republishes a document with only its “updated as on” date changed after a minor amendment.

B. PDF/text extraction

Layout-aware text extraction recovers the document’s textual content while preserving paragraph and sub-paragraph numbering, headings, and table structure, since RBI documents rely on consistent structural numbering to convey clause hierarchy. We follow the finding, established for layout-aware document pre-training by Xu et al. [23], that treating a regulatory document as a flat token stream and discarding positional and typographic structure measurably degrades downstream extraction; the extraction step therefore preserves each text span’s page position, indentation level, and any enclosing table cell coordinates, rather than emitting a single concatenated string. RBI Master Directions typically nest content four levels deep – Chapter, Part, Paragraph, and sub-paragraph (for example, “Chapter III, Para 7, sub-para (3)(ii)”) – and the extraction step reconstructs this nesting explicitly rather than inferring it later from indentation heuristics applied to already-flattened text.

C. Document cleaning and normalization

Extracted text is normalized to remove page-header and page-footer artifacts (RBI documents place a running header with the circular reference number and a footer with the page number and department name on every page, both of which would otherwise be mis-segmented as body content), repair hyphenation introduced by line wrapping, and standardize whitespace, without altering the substantive wording of the regulatory text. Footnote markers and their corresponding footnote text – common in RBI Master Directions for citing an amending circular – are detected and re-attached to the clause they annotate rather than left as an orphaned trailing block.

D. Clause segmentation

Segmentation is performed deterministically using RBI’s regular structural markers – paragraph and sub-paragraph numbering conventions used consistently within a given document type – rather than through an LLM, since these markers are reliable enough for rule-based parsing and this avoids spending model calls on a task that does not require semantic understanding. A segmentation rule set (regular-expression patterns matched against the numbering conventions of Master Directions, Master Circulars, and standalone circulars separately, since the three document types number their content differently) produces a candidate clause boundary set, which is then validated against the document’s own table of contents (where present) as a consistency check: a document whose

detected clause count disagrees with its table of contents by more than a small tolerance is flagged for manual review rather than silently ingested with a possibly broken clause structure.

E. Regulatory definition extraction

Defined terms (for example, a Master Direction’s “Definitions” clause, typically numbered as an early paragraph such as “Para 3 – Definitions”) are extracted and linked to every subsequent clause in which the term is used, so that downstream simplification and question answering can resolve a term to its governing definition rather than relying on the reader’s prior familiarity with it. Term linking uses exact and near-exact string matching against the extracted definition list within the scope of the document (and, where cross-reference handling below has resolved a link, within cross-referenced documents), rather than a general-purpose named-entity recognizer, because RBI definitions are self-contained and enumerated rather than requiring open-domain entity recognition.

F. Rule and obligation identification

Each segmented clause is examined to identify whether it states an obligation (“banks shall ...”), a prohibition (“no bank shall ...”), a permission (“banks may ...”), a definition, or a purely explanatory or preambular statement, forming the basis for the rule extraction pipeline detailed in Section VI. This classification is performed by an LLM prompted with a small set of in-context examples covering each of the five categories together with the deontic-verb patterns (shall, shall not, may, is required to, is exempt from) RBI text uses to express them, with its output constrained to one of the five labels rather than free text.

G. Regulatory category classification

Clauses are classified into operational categories such as Know Your Customer (KYC), anti-money-laundering (AML), lending, transaction reporting, governance, and audit, so that both the human-facing search interface and the compliance rule repository can be organized and filtered by category. A clause may receive more than one category label (for example, a reporting obligation attached to a KYC requirement is labeled both KYC and transaction reporting), since RBI requirements frequently span more than one operational area and forcing a single label would lose that overlap.

H. Simplification of regulatory language

For each clause, the module generates a simplified natural-language restatement intended for a non-legal reader, produced under an LLM prompting strategy constrained to preserve the clause’s substantive conditions and obligations rather than merely shortening its wording; the original clause text remains the authoritative source and is always presented alongside the simplification, and the interface never presents the simplification as a substitute the reader cannot trace back to source. Simplification quality is evaluated along two axes that can trade off against each other – how much simpler the text

has become, and how much of the original meaning survives – using the metrics discussed in Section XIII.C, since a simplification that is easy to read but drops an exception clause is not actually useful to a compliance officer.

I. Structured regulatory knowledge representation

The output of clause-level processing is stored as the typed rule representation defined in Section VI, which is expressive enough to capture quantitative, Boolean, conditional, temporal, qualitative, relational, and multi-entity content within a single schema, so that regulation comprehension and compliance checking read from the same object rather than two independently maintained representations of the same clause.

J. Cross-reference handling

References from one clause to another (“as specified in paragraph 4 above”), or to a separate Master Direction or circular (“in terms of the Master Direction on KYC dated ...”), are resolved to the target clause identifier where the target has itself been ingested, so that a rule’s applicability or exception conditions can be traced across documents rather than treated as opaque text. Where the referenced document has not yet been ingested, the reference is retained as an unresolved pointer (source document name and, where extractable, a date) rather than silently discarded, so that it can be resolved automatically once that document is ingested.

K. Amendment and supersession handling

When a newly ingested document amends or supersedes a previously ingested clause, the module records a supersession relationship (Section VI, field `supersedes`) rather than silently overwriting the prior version, consistent with the versioning approach described below. Because RBI amendments are frequently partial – a circular may amend a single subparagraph of a much longer Master Direction – the module records supersession at clause granularity, not document granularity, so that the ninety-nine unamended clauses of a Master Direction are not needlessly revisioned when only one clause changes.

L. Version and effective-date management

Every extracted rule records an effective-from date and, where applicable, an effective-to date, so that a query directed at a specific point in time can be resolved against the regulation as it stood on that date rather than only against the most recent version. This is what makes the cross-version benchmark cases of Section XII meaningful: a compliance query dated before an amendment’s effective date must be answered against the pre-amendment rule, and the same evidence evaluated after that date must be answered against the amended rule, using the same underlying versioning metadata rather than a special-cased comparison.

M. Retrieval mechanism

Clause and definition text is embedded and indexed for semantic retrieval, so that a natural-language query can be matched to the most relevant clauses even when the query’s wording does not lexically match the regulatory text – for example, a user asking about “cash transaction limits” should retrieve a clause phrased in terms of “currency transaction reports” even without a shared keyword. Retrieval quality is measured directly (Section XIII.B) rather than assumed, since a citation-constrained answer generator (below) can only be as accurate as the clauses it is given to work with.

N. Natural-language question answering

A user question is answered by retrieving the most relevant clauses, then generating a response conditioned on the retrieved text under a citation-constrained prompting strategy, following the retrieval-augmented generation pattern introduced by Lewis et al. [24] and discussed further in Section III-C. The generator is instructed to answer using only the retrieved clauses, to quote or closely paraphrase the operative condition rather than the surrounding preamble, and to state explicitly when the retrieved clauses do not fully answer the question rather than filling the gap from unretrieved background knowledge.

O. Source and clause citation

Every answer returned by the question-answering interface cites the specific RBI circular, clause identifier, and, where relevant, the regulation version, so that the user can verify the answer against the original source rather than relying on the system’s restatement alone. This citation is stored as structured metadata (document id, clause id, version) alongside the generated text, not inferred after the fact by pattern-matching the answer against the corpus, since the latter would only detect citation errors rather than prevent them.

P. Hallucination mitigation

Beyond citation, the module constrains answer generation to the retrieved passages and declines to answer, rather than speculating, when retrieval confidence for a query falls below a configured threshold; this is the same evidence-grounding principle applied to compliance decisions in Section IX, extended to the human-facing interface. We adopt the faithfulness and answer-relevance decomposition used in the RAGAs evaluation framework [25] as the basis for measuring how well this constraint holds in practice (Section XIII.C), and take the broader hallucination-mitigation taxonomy of Ji et al. [26] – of which retrieval grounding, citation, and abstention are three specific instances – as the framing under which this design choice is made, rather than treating grounding as a property that follows automatically from using retrieval-augmented generation at all.

Q. Reuse by the compliance checking module

The structured regulatory knowledge produced by this module – typed rule objects, clause citations, category labels, and version and effective-date metadata – is not regenerated for Adaptive Compliance Checking; it is read directly from the shared regulatory knowledge base described in Section X. This reuse is the basis of the paper’s system-level integration contribution.

R. Worked example

To make the module’s behavior concrete, consider RBI’s Master Direction on Know Your Customer, paragraph 16, which requires enhanced due diligence for customers assessed as high risk. Ingestion (Section V.A–B) recovers the paragraph’s text together with its position under Chapter V; segmentation (Section V.D) isolates it as a single clause distinct from the surrounding paragraphs 15 and 17; definition linking (Section V.E) resolves the term “high risk customer” to its governing definition in paragraph 3; rule identification (Section V.F) labels the clause as an obligation; category classification (Section V.G) labels it KYC; and rule extraction (Section VI) produces a category-E (qualitative) rule object with `requires_semantic_interpretation = true`, since “enhanced due diligence” is not reducible to a numeric or Boolean test. A subsequent user question, “what extra checks are needed for a high-risk customer,” retrieves this clause (Section V.M) and receives a simplified, citation-bearing answer (Section V.I, N–O) referencing Master Direction – KYC, paragraph 16. The same rule object, unchanged, is later read by Adaptive Compliance Checking (Section VII) when evaluating a specific bank’s onboarding file for a high-risk customer, and because `requires_semantic_interpretation = true`, the router (Section VIII) directs that evaluation to the retrieval-augmented LLM path rather than the rule engine.

VI. REGULATORY RULE REPRESENTATION

A flat (condition, operator, threshold) triple, while sufficient for simple quantitative rules, cannot represent conditional (“if ...and ...then ...unless”), temporal, or exception-bearing clauses without ad hoc extensions. Every extracted rule is instead represented as a typed object with a nested logic block, summarized in Table III and given in full in Appendix A.

A. Requirement category taxonomy

Extracted clauses are classified into seven categories, summarized in Table IV. Categories A through D, together with the Boolean-evidence subset of category E and the non-graph cases of category G, are routed preferentially to the deterministic rule engine when evidence is complete; the remainder of category E and the unresolved cases of category F are routed to retrieval-grounded LLM reasoning (Section VIII).

VII. ADAPTIVE COMPLIANCE CHECKING MODULE

This module addresses the supervisory side of the problem: given a bank’s submitted evidence and the applicable RBI regulatory requirements, it produces a traceable compliance decision. As with Regulation Comprehension (Section V), we describe each processing step in enough detail to support independent implementation and evaluation, since this module – specifically its routing component (Section VIII) – carries the paper’s primary contribution and is evaluated with the same rigor as Module 1 (Section XII).

A. Bank data ingestion

Banks periodically submit audit reports, compliance filings, and operational records in structured (CSV/JSON) or text-extractable PDF form, consistent with the scope decision in Section IV.A. A submission is tagged with the submitting entity’s identifier, the reporting period it covers, and the RBI return or report type it corresponds to (for example, a DSB return or an internal audit report), so that evidence from different reporting periods for the same entity is never silently merged.

B. Data normalization

Submitted data is mapped to a canonical evidence schema so that fields referenced by a rule’s antecedent (for example, “single-counterparty exposure as a fraction of eligible capital”) can be located consistently regardless of the bank’s original reporting format. Because different banks’ internal systems export the same underlying quantity under different column names or units (for example, exposure reported in absolute currency versus already expressed as a ratio), the normalization step also performs unit reconciliation against the unit recorded in the corresponding rule’s schema (Section VI, field `unit`), flagging a mismatch rather than silently coercing it.

C. Evidence extraction

Where a submission is a text-extractable document rather than structured data, relevant evidence values and supporting statements are extracted and linked to the canonical schema field they populate, using the same layout-aware extraction approach used for regulatory documents (Section V.B), since bank audit reports share the same practical difficulty of mixing narrative text with embedded tables.

D. Evidence completeness

For each rule under evaluation, the module computes the fraction of the rule’s required evidence fields that are present and non-null; this fraction, `evidence_completeness`, is a primary feature of the router described in Section VIII. Completeness is computed per rule, not per submission, because a single bank submission can be fully sufficient for one rule and insufficient for another that requires an additional field the submission does not contain.

TABLE III
CORE FIELDS OF THE TYPED RULE REPRESENTATION (ABRIDGED; FULL SCHEMA IN APPENDIX A).

Field	Type	Purpose
rule_id	string	Stable identifier, includes source document and clause
source_document / clause_id	string	Traceability to the originating Master Direction / circular
rule_type	enum (A–G)	Quantitative, Boolean, conditional, temporal, qualitative, relational, multi-entity
applicability	object	Entity types and conditions under which the rule applies
logic.antecedent / consequent / exceptions	nested object	Composable Boolean, conditional, and quantitative logic in one schema
temporal	object	effective_from, effective_to, reporting_frequency
requires_semantic_interpretation	boolean	Set at extraction time; primary router feature (Section VIII)
confidence / human_verified	float / boolean	Extraction-quality metadata, independent of downstream decision confidence
version / supersedes	int / rule_id	Versioning and amendment tracking (Section V)

TABLE IV
REQUIREMENT CATEGORY TAXONOMY AND EVALUATED-SCOPE STATUS.

Category	Description	Preferred reasoning modality	Status
A – Quantitative	Thresholds, ratios, numerical limits	Rule engine (evidence complete)	Implemented / evaluated
B – Boolean / procedural	Existence checks (e.g., board approval, documentation)	Rule engine (evidence complete)	Implemented / evaluated
C – Conditional	If ... and ... then ... unless clauses	Rule engine (evidence complete)	Implemented / evaluated
D – Temporal	Deadlines, periodic reporting, date-conditioned applicability	Rule engine (evidence complete)	Implemented / evaluated
E – Qualitative / semantic	“Adequate due diligence,” “reasonable measures”	Retrieval-augmented LLM	Implemented / evaluated
F – Relational	Parent/subsidiary, related-party requirements	Retrieval-augmented LLM (non-graph cases)	Partially implemented
G – Multi-entity / connected-counterparty	Exposure aggregation across related entities	Rule engine (proposed)	Not implemented; future work (Section XX)

E. Matching bank evidence with regulatory requirements

Applicable rules for a given bank and reporting period are selected from the shared regulatory knowledge base according to the rule’s applicability conditions (entity type, bank category, and any stated conditions), read directly from the knowledge base populated by Regulation Comprehension (Section X) rather than a separately maintained rule list.

F. Rule applicability

A rule is considered applicable to a given evidence set only if the entity and conditions recorded in its applicability object are satisfied; inapplicable rules are excluded from the compliance evaluation for that entity rather than evaluated and discarded post hoc, so that an inapplicable rule can never surface as a false NON_COMPLIANT finding.

G. Compliance decision

For each applicable rule, the routed reasoning modality (Section VIII) produces one of five decisions, summarized in Section VII.K below.

H. Evidence traceability

Every decision is stored together with the evidence trace τ defined in Section II, linking the decision to the specific rule, clause, and evidence fields that produced it (Section IX), so that a reviewer can inspect exactly which submitted values were consulted and which regulatory clause they were tested against.

I. Corrective recommendations

Where a decision is NON_COMPLIANT, the system generates a recommended corrective action derived from the rule’s stated requirement (for example, “reduce single-counterparty exposure to below 20% of eligible capital or document an

applicable exemption”), distinguished in the report from the compliance decision itself so that a recommendation is never presented as though it were a binding regulatory determination.

J. Audit-ready report generation

A template-based report is generated referencing the rule identifier, clause, evidence, reasoning modality used, decision, explanation, and any corrective recommendation, described further in Section IX. The report format is held constant regardless of which reasoning modality produced the underlying decision, so that a reviewer reading the report cannot distinguish a rule-engine decision from an LLM-produced one by report structure alone; the reasoning-modality field itself, not the report’s shape, is what discloses this.

K. Compliance decision outputs

The system returns one of five decisions for each (rule, evidence) query:

- **COMPLIANT**: the available evidence satisfies the rule’s stated requirement.
- **NON_COMPLIANT**: the available evidence affirmatively fails the rule’s stated requirement.
- **INSUFFICIENT_DATA**: required evidence fields are missing, and the system declines to infer a decision from incomplete evidence.
- **AMBIGUOUS**: the available evidence does not unambiguously satisfy or violate the rule, typically arising for qualitative (category E) requirements or where evidence sources disagree.
- **REQUIRES_HUMAN_REVIEW**: the router (Section VIII) has determined that no automated modality can provide a sufficiently defensible decision, and the case is escalated.

L. Worked example

Continuing the example of Section V-R: suppose Bank X submits an onboarding file for a customer classified, under the definition resolved during comprehension, as high risk. Data normalization (Section VII.B) maps the submission’s fields into the canonical schema; evidence completeness (Section VII.D) is computed against the category-E rule’s required fields (in this case, narrative fields describing the due-diligence steps taken, which do not reduce to a numeric completeness check in the same way a quantitative rule’s fields would); rule applicability (Section VII.F) confirms the rule applies to Bank X’s entity type; and because the rule’s `requires_semantic_interpretation` flag is true (Section V-R), the router (Section VIII) sends the query to the retrieval-augmented LLM path rather than the rule engine, which reasons over the submitted due-diligence narrative against the retrieved clause text and returns COMPLIANT, AMBIGUOUS, or NON_COMPLIANT together with an evidence trace citing both the submitted narrative field and Master Direction – KYC, paragraph 16. A second, unrelated query for the same bank – whether its single-counterparty exposure evidence satisfies the quantitative large-exposure threshold of Appendix A – has `rule_engine_applicability` = 1 and, if evidence is complete, is sent to the deterministic rule engine under the hard override of Section VIII.F regardless of the qualitative case just described, illustrating within a single bank’s submission how the same pipeline routes two different requirement types to two different reasoning modalities.

VIII. ADAPTIVE REASONING-MODALITY ROUTER

This section presents the paper’s primary contribution: a routing mechanism that selects a reasoning modality per (rule, evidence) query under an explicit cost objective, evaluated against baselines drawn from the LLM-routing literature’s own evaluation standard (Section III).

A. Why routing is necessary

Section II established that the system’s overall decision function decomposes as $f(r_i, E) = f_{\rho(r_i, E)}(r_i, E)$, where the router $\rho(r_i, E) \rightarrow m$ selects a reasoning modality m and a modality-specific reasoner $f_m(r_i, E) \rightarrow (d, \tau)$ produces the decision and evidence trace. Formally,

$$\rho(r_i, E) \rightarrow m, \quad m \in \{\text{RULE_ENGINE}, \text{RAG_LLM}, \text{HUMAN_REVIEW}\}. \quad (3)$$

The three candidate modalities differ not only in cost and latency but in the correctness guarantee they can offer: $f_{\text{RULE_ENGINE}}$ is deterministic and, for a rule with complete, well-typed evidence, correct by construction relative to the extracted rule logic; $f_{\text{RAG_LLM}}$ is grounded but not deterministically guaranteed; $f_{\text{HUMAN_REVIEW}}$ defers the decision entirely. Routing is necessary because no single modality is appropriate across all seven requirement categories of Table IV: applying the rule engine to a qualitative requirement is not possible without discarding the requirement’s semantic content, and applying the LLM to a fully specified numeric requirement introduces avoidable risk and cost.

B. Feature representation

The router operates on a feature vector computed per (rule, evidence) query, summarized in Table V.

C. Objective

For a candidate route m and query (r_i, E) , define an expected cost

$$\begin{aligned} \text{Cost}(m) = & \alpha \cdot P(\text{error} \mid m, \text{features}) \cdot \text{RiskWeight}(r_i) \\ & + \beta \cdot \text{LLMTokenCost}(m) \\ & + \gamma \cdot \text{Latency}(m), \end{aligned} \quad (4)$$

subject to a validation-set constraint that overall compliance accuracy remains above a threshold τ_{acc} (for example, $\tau_{\text{acc}} = 0.95$, tuned on held-out data, not on the test benchmark). The router selects the modality m minimizing $\text{Cost}(m)$ subject to this constraint. The weights α , β , and γ are tuned by grid search on a validation split, and the intended reporting format is a full accuracy–cost–latency trade-off surface (Section XVI) rather than a single operating point, consistent with how the LLM-routing literature reports Pareto frontiers rather than single aggregate numbers.

D. Candidate routers compared

- **R1 – Static / rule-first:** always attempt the rule engine; fall back to the retrieval-augmented LLM only if required structured inputs are missing.
- **R2 – Confidence-threshold:** route by a single, manually tuned threshold on `retrieval_confidence` or `rule_engine_applicability`. This is the routing strategy most systems default to in practice, and is the baseline the proposed router must beat, not merely match.
- **R3 – Fixed hybrid:** a manually authored decision table mapping `rule_type` to a fixed route, with no learned or per-query adaptation (corresponds to Baseline B4, Section XIV).
- **R4 – Proposed adaptive router:** a learned classifier estimating $P(\text{error} \mid m, \text{features})$ per candidate route, combined with the cost objective of Eq. (4), optionally implemented as a contextual-bandit policy over the three routes.

Algorithm 1: Adaptive reasoning-modality routing

Require: rule r , evidence E , trained router model M

Ensure: selected reasoning modality m^*

```

1: features  $\leftarrow$  EXTRACTFEATURES( $r, E$ ) {Table V}
2: if features.evidence_completeness < MIN_EVIDENCE_THRESHOLD then
3:   return HUMAN_REVIEW
4: end if
5: for  $m \in \{\text{RULE\_ENGINE}, \text{RAG\_LLM}, \text{HUMAN\_REVIEW}\}$  do
6:    $p_{\text{error}} \leftarrow M.\text{PREDICTERRORPROB}(\text{features}, m)$ 
7:    $\text{cost}[m] \leftarrow \alpha \cdot p_{\text{error}} \cdot r.\text{risk\_weight}$ 
8:      $+ \beta \cdot \text{ESTIMATEDLLMCOST}(m, E)$ 
9:      $+ \gamma \cdot \text{ESTIMATEDLATENCY}(m)$ 
```

TABLE V
ROUTER FEATURE VECTOR.

Feature	Definition
rule_type	One-hot encoding of categories A–G (Table IV)
requires_semantic_interpretation	Extraction-time flag from the rule schema
structured_field_availability	Fraction of the rule’s required evidence fields present in structured bank data
retrieval_confidence	Top- k similarity score from the vector store for the RAG path
rule_engine_applicability	1 if the rule’s logic is fully deterministic and all required inputs are numeric/Boolean and present; else 0
evidence_completeness	Fraction of non-null required evidence fields, over all fields
historical_router_accuracy[route]	Rolling validation-set accuracy of each candidate route for similar rule_type
risk_level	Rule-level risk annotation (e.g., capital adequacy = high; documentation = low)

```

10: end for
11:  $m_{\text{best}} \leftarrow \arg \min_m \text{cost}[m]$ 
12: if features.rule_engine_applicability = 1 and
    features.evidence_completeness = 1.0 then
13:   return RULE_ENGINE {Hard override,
    Section VIII.F}
14: end if
15: return  $m_{\text{best}}$ 

```

Algorithm 1 presents the routing procedure. Cases with evidence completeness below a minimum threshold are escalated to human review before the cost objective is evaluated, since no automated modality can be trusted to produce a defensible decision from insufficient evidence. Otherwise, the expected cost of each candidate modality is computed and the minimizing modality is selected, subject to the hard override described next.

E. The hard override and its role in the correctness claim

The router’s learned component decides among genuinely ambiguous cases; it is not what guarantees determinism for the fully specified numeric and Boolean subset of rules. That guarantee is architectural rather than statistical: any rule with `rule_engine_applicability = 1` and `evidence_completeness = 1.0` is sent to the deterministic rule engine unconditionally, regardless of what the learned router would otherwise predict. This design decision lets the system state a correctness property independent of router accuracy – one hundred percent of category A/B/C/D rules with complete evidence are answered deterministically, by construction – rather than a weaker statistical claim about how often the router happens to choose correctly. This is also the specific respect in which reasoning-modality routing differs from LLM-to-LLM routing (Section III): the latter has no analogous notion of a modality that is correct by construction, because every candidate in that literature is itself a probabilistic language model.

IX. EVIDENCE-GROUNDED COMPLIANCE AND EXPLAINABILITY

Every decision produced by the system, whether generated by the rule engine, the retrieval-augmented LLM, or returned as a human-review escalation, is linked to: the source regulatory document; the specific clause; the rule identifier; the regulation version in force at the relevant date; the bank evidence fields consulted; the reasoning modality used; the

resulting decision; a natural-language explanation; and, where applicable, a recommended corrective action. This evidence trace τ (Section II) is stored alongside the decision rather than reconstructed after the fact, so that an auditor can verify a decision without re-running the pipeline.

Grounding serves two purposes that are worth distinguishing. First, at generation time, constraining the retrieval-augmented LLM to cite only retrieved clauses (Section V) reduces the opportunity for the model to state a rule or threshold that does not exist in the source corpus. Second, at audit time, the stored trace allows a human reviewer to check the citation against the source document directly, independent of whether the generating model is later updated or becomes unavailable. We treat evidence-citation accuracy – whether a cited rule identifier and clause genuinely support the stated decision – and hallucination rate – the fraction of decisions citing a non-existent or incorrect clause – as measurable quantities (Section XV) rather than as an architectural property assumed to hold by design; Section XVI reports these as placeholders pending evaluation.

X. END-TO-END INTEGRATION OF BOTH MODULES

Regulation Comprehension and Adaptive Compliance Checking are presented in Sections V and VII as separate modules because they serve different users and produce different outputs, but they are not two disconnected applications: they operate over the same regulatory knowledge base, and the structured rule representation produced once during comprehension is read, not regenerated, by the compliance module. The dependency runs in one direction only, illustrated below:

RBI regulation $\rightarrow$ comprehension $\rightarrow$ structured rules $\rightarrow$
searchable regulatory knowledge $\rightarrow$ compliance evaluation.

The same extracted regulatory representation therefore serves two purposes: a human-facing purpose, in which a compliance officer or auditor queries the knowledge base through natural language and receives a cited, simplified answer (Section V); and a machine-facing purpose, in which the same typed rule objects are consumed by the router and reasoning engines of Section VIII to evaluate bank evidence.

This sharing has two practical consequences that would not hold if the two modules maintained independent extraction pipelines. First, a correction to a rule’s extracted representation – for example, following human verification (Section V) or

an amendment update – propagates automatically to both the question-answering interface and the compliance decision engine, rather than requiring the correction to be made twice and risking the two representations drifting apart. Second, the version and effective-date metadata recorded once during comprehension (Section V) is the same metadata that resolves cross-version compliance test cases (Section XII), so that a query about “what does the current regulation require” and a compliance check against “the regulation in force on a given date” are answered from a single consistent source rather than two representations that could disagree.

We are explicit that this integration, by itself, is not presented as the paper’s primary novel contribution: connecting a comprehension module to a downstream reasoning module is a natural systems-engineering choice rather than a new technique. It is presented as a system-level contribution that makes the primary contribution – adaptive reasoning-modality routing (Section VIII) – practically deployable within a single coherent regulatory intelligence workflow, rather than as a standalone research claim.

XI. SYSTEM ARCHITECTURE AND IMPLEMENTATION

This section describes the implementation supporting the architecture of Section IV, distinguishing components that are implemented from components that remain proposed or future work.

A. Relational and vector storage

Structured rules, rule versions, clauses, source documents, applicability conditions, and compliance results are normalized into a relational schema in PostgreSQL, following the fields defined in Section VI and Appendix A. Clause and definition embeddings for semantic retrieval are indexed using pgvector, co-located with the relational store rather than a standalone vector database, given a corpus scale (thousands, not billions, of vectors) consistent with the benchmark scale discussed in Section XII.

B. Rule engine

The deterministic reasoning modality executes the logic block of each rule object directly using a JSON-Logic-style rule engine, avoiding both a Java-virtual-machine-based engine and a custom domain-specific language that would itself require separate validation.

C. LLM with structured output and function calling

Two uses of a large language model are implemented: schema-constrained extraction (Section V), in which the model is constrained to emit valid instances of the schema of Section VI, and retrieval-augmented reasoning (Sections V and VIII), in which the model reasons over retrieved clauses under a citation-constrained prompting strategy. Model identity and version are to be pinned and recorded (Section XII) rather than left unspecified, since extraction and routing results are expected to shift if the underlying model is updated.

D. Embeddings and retrieval

Clause and definition text is embedded with a fixed, recorded embedding model (Section XII) and retrieved by vector similarity, feeding both the question-answering interface of Section V and the retrieval-augmented reasoning path of Section VIII.

E. Regulatory document and evidence processing

Document ingestion (Section V) and bank-evidence ingestion (Section VII) use format-specific parsers – layout-aware PDF/HTML extraction for regulatory documents, and CSV/JSON/PDF-text parsers mapped to a canonical schema for bank evidence – rather than a single generic parser, since the two input types differ substantially in structure and reliability.

F. Offline versus query-time processing

Regulatory-side processing (document ingestion, clause segmentation, rule extraction, and knowledge-base population) executes offline or asynchronously as new or amended RBI documents are ingested. Compliance-side processing (evidence ingestion, routing, reasoning, and decision assignment) executes at query time, against the regulatory knowledge base already populated offline. This separation keeps the latency-sensitive query-time path from depending on document-processing latency.

G. Implemented versus proposed components

The relational and vector storage layers, the deterministic rule engine, schema-constrained extraction, retrieval-augmented question answering and reasoning, and the router’s feature extraction (Table V) are implemented components of the architecture as described. The learned cost-minimizing router policy of Section VIII (as distinct from its feature representation and hard override, both of which are architectural rather than learned) and the connected-counterparty graph reasoning of Section VI (category G) are, respectively, planned for evaluation (Section XII) and proposed future work (Section XX); we do not present either as already validated.

XII. EXPERIMENTAL METHODOLOGY

Consistent with treating both modules as independently measured contributions (Section V), this section presents two methodologies side by side: Section XII.A–F covers Adaptive Compliance Checking, and Section XIII covers Regulation Comprehension. Both follow the same construction discipline – systematically derived from source material rather than authored independently of it, with disclosed provenance – so that the two sets of results in Section XVI are comparable in rigor even though they measure different capabilities.

A. Benchmark construction (compliance checking)

Because bank operational data is confidential and cannot be obtained for academic use (Section XIX), we construct a synthetic benchmark systematically derived from the extracted rule corpus (Section VI) rather than generated independently

of it, so that every test case traces to a specific rule_id and its source clause. This follows recent calls in the financial-LLM-evaluation literature for benchmarks that go beyond static question-answer pairs to include boundary, adversarial, and escalation scenarios [17]–[19].

B. Case categories (compliance checking)

Table VI summarizes the nine case-difficulty categories used to construct the benchmark.

C. Case schema and provenance (compliance checking)

Each case is stored as a record with fields case_id, entity_info, rule_id, input_data, relevant_documents, expected_decision, expected_reason, difficulty_level, case_type, regulation_version, and generation_method. The generation_method field (template, LLM-perturbation, or human-authored) is retained specifically so the benchmark’s provenance can be reported honestly, since recent work explicitly asks synthetic-benchmark papers to disclose this rather than presenting synthetic data as though it were equivalent to naturally occurring data [17], [19].

D. Intended scale (compliance checking)

[PLACEHOLDER – to be finalized once extraction, Section V, is complete]: the benchmark targets 500–1,500 cases drawn from 40–80 distinct rules across 3–5 RBI Master Directions, deliberately bounded rather than attempting exhaustive coverage of RBI regulation, so that per-case construction quality, and, where feasible, human review, remains tractable for a single-team research project.

E. Training, validation, and test separation

The compliance benchmark is intended to be split into training, validation, and test partitions such that the learned router policy (Section VIII) and its cost weights α , β , γ are tuned only on the training and validation partitions, with the accuracy constraint τ_{acc} (Section VIII) enforced on the validation partition, and all reported figures (Section XVI) computed on the held-out test partition. The comprehension benchmark (Section XIII) is split identically in spirit – by source document, so that no clause from a test-partition document is ever seen in extraction-prompt few-shot examples tuned on the training partition.

F. Implementation details for reproducibility

The following details are to be recorded once the pipeline is run, and are left as explicit placeholders here rather than populated with unverified values: the relational schema is implemented in PostgreSQL, normalized into rules, rule_versions, clauses, documents, applicability_conditions, and compliance_results tables (Section XI); the vector store is pgvector, co-located with the relational store; the rule engine executes the logic block of each rule object directly via a JSON-Logic-style engine; the large language model used for extraction and retrieval-augmented reasoning is [MODEL NAME AND VERSION TO BE RECORDED HERE]; the embedding model is [EMBEDDING MODEL TO BE RECORDED

HERE]; and the rule-extraction verification sample size is [N CLAUSES TO BE RECORDED HERE], manually verified against source text across [K] Master Directions.

XIII. EXPERIMENTAL METHODOLOGY FOR REGULATION COMPREHENSION

Module 1 is evaluated on its own held-out benchmark, distinct from the compliance-checking benchmark of Section XII, because it measures a different capability – faithful extraction, retrieval, and explanation of what a regulation says, independent of any downstream compliance judgment.

A. Benchmark construction (regulation comprehension)

The comprehension benchmark is constructed from the same ingested RBI Master Directions and circulars used to populate the shared regulatory knowledge base (Section X), so that every comprehension test case, like every compliance test case, traces to a specific source clause. Three parallel sub-benchmarks are constructed: an *extraction* sub-benchmark of clause-level ground truth for segmentation, definition linking, obligation typing, and category labels; a *question-answering* sub-benchmark of natural-language questions paired with a gold answer, gold supporting clause(s), and gold regulation version; and a *simplification* sub-benchmark of clause/simplification pairs with expert-authored reference simplifications. This mirrors, at a construction level, how expert-annotated legal datasets such as CUAD [28] and numerical financial-QA datasets such as FinQA [29] pair source documents with span- or answer-level ground truth rather than treating the source text and the ground truth as independently collected.

B. Extraction sub-benchmark

Ground truth for clause segmentation, definition extraction, obligation-type labels (obligation/prohibition/permission/definition/explanatory, Section V.F), and category labels (Section V.G) is produced by manual annotation of a disclosed sample of ingested documents, following the same discipline used for rule-extraction verification in Section XII.F: sample size and source documents are to be recorded rather than left implicit, and annotation is performed independently of, and prior to, running the extraction pipeline against the same clauses, so that the pipeline’s own output cannot leak into its evaluation.

C. Question-answering sub-benchmark

Natural-language questions are constructed in three ways, each labeled by its generation_method field analogous to the compliance benchmark (Section XII.C): template questions generated mechanically from a rule’s applicability and antecedent fields (“what is the exposure limit for a single counterparty”); paraphrase questions that reword a template question without changing its target clause, testing retrieval robustness to lexical variation; and multi-clause questions whose answer requires combining two or more clauses (for example, a general rule and a stated exception), testing whether

TABLE VI
BENCHMARK CASE CATEGORIES AND CONSTRUCTION METHOD (ADAPTIVE COMPLIANCE CHECKING).

Case type	Construction method
Compliant / Non-compliant	Evidence values sampled clearly on the satisfying / violating side of the rule’s threshold or condition.
Boundary	Evidence value exactly equal to the threshold; forces an explicit, documented convention for $\leq$ versus $<$ per rule.
Near-boundary	Evidence within $\pm 1\text{--}5\%$ of threshold, testing sensitivity and robustness rather than gross correctness.
Ambiguous	Evidence for qualitative (category E) rules that does not unambiguously satisfy or violate the stated language; expected_decision = AMBIGUOUS, not a forced binary label.
Missing-data	Required evidence fields deliberately withheld; expected_decision = INSUFFICIENT_DATA, testing whether a system fabricates a decision under missing evidence.
Adversarial	Evidence phrased to appear compliant in qualitative language while numerically failing (or vice versa), testing whether surface phrasing can override a deterministic check that should dominate.
Contradictory	Two evidence sources disagree; expected_decision = AMBIGUOUS or REQUIRES_HUMAN_REVIEW with both sources cited.
Cross-version	Identical fact pattern evaluated against two rule_versions (pre-/post-amendment), with different expected outcomes, testing versioning and effective-date logic.

retrieval returns a sufficient evidence set rather than only the single best-matching clause. Each question records a gold answer, the gold supporting clause identifier(s), and, for cross-version questions, the regulation version the question is posed against.

D. Simplification sub-benchmark

A subset of extracted clauses, stratified across the seven requirement categories of Table IV so that simplification quality is not measured only on the easiest (quantitative) clause types, is paired with an expert-authored reference simplification. Reference simplifications are authored independently by someone other than whoever tunes the simplification prompt, to avoid the reference set implicitly encoding the system’s own phrasing preferences.

E. Intended scale (regulation comprehension)

[PLACEHOLDER – to be finalized once extraction is complete]: the extraction sub-benchmark targets full-coverage annotation of the same 3–5 Master Directions used for the compliance benchmark (Section XII.D); the question-answering sub-benchmark targets 300–600 questions (template, paraphrase, and multi-clause combined) with a target ratio of roughly 2:2:1 across those three types; and the simplification sub-benchmark targets 100–200 clause/reference pairs, stratified across categories A–G.

F. Evaluation protocol

Extraction metrics (Section XV-B.A) are computed by comparing pipeline output against the extraction sub-benchmark’s manual annotations. Retrieval and question-answering metrics (Section XV-B.B) are computed against the question-answering sub-benchmark, with retrieval evaluated independently of generation (whether the correct clause appears in the top- k retrieved set) before end-to-end answer quality is evaluated, so that a retrieval failure and a generation failure are never conflated into a single opaque number. Simplification metrics (Section XV-B.C) are computed against the simplification sub-benchmark. As with the compliance benchmark, no

model or prompt tuning is permitted against the held-out test partition of any of the three sub-benchmarks.

XIV. BASELINES

A. Baselines for Adaptive Compliance Checking

Table VII summarizes the four baselines and the proposed system, each isolating a different design choice.

B. Baselines for Regulation Comprehension

Module 1 is compared against three baselines representing progressively richer comprehension architectures, summarized in Table VIII, so that the contribution of RBI-specific structuring (Section V) can be isolated from the contribution of retrieval-augmented generation in general [24].

XV. EVALUATION METRICS

A. Metrics for Adaptive Compliance Checking

Metrics are grouped by research question rather than reported as an undifferentiated pool.

RQ1/RQ4 (compliance accuracy and grounding): accuracy, precision, recall, F1, false-positive rate, false-negative rate, evidence-citation accuracy (whether the cited rule_id and clause actually support the decision), and hallucination rate (the fraction of decisions citing a non-existent or incorrect clause), each reported overall and stratified by case_type (Table VI). For a decision set with true positives TP , false positives FP , true negatives TN , and false negatives FN ,

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (5)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (6)$$

RQ2 (routing efficiency): the accuracy–cost–latency Pareto frontier across routers R1–R4 (Section VIII); routing accuracy against a hand-labeled “correct route” reference set; and the fraction of LLM calls avoided relative to B2/B3.

RQ3 (extraction quality): entity, operator, threshold, and unit extraction precision, recall, and F1, and rule executability rate, all broken down by requirement category A–G (Table IV).

TABLE VII
BASELINES COMPARED AGAINST THE PROPOSED SYSTEM (ADAPTIVE COMPLIANCE CHECKING).

Baseline	Description	Known limitation
B1 – Rule-only	Structured rules executed by the deterministic engine only.	No coverage of qualitative (category E) requirements; brittle to missing or unstructured evidence.
B2 – LLM-only	Raw bank data and regulatory text passed directly to the LLM with no retrieval or rule-engine grounding.	No grounding guarantee; expected to be most exposed to hallucination on adversarial and missing-data cases.
B3 – RAG+LLM	Vector retrieval over regulatory clauses feeding LLM reasoning, with citation.	Adds grounding over B2 but no deterministic guarantee on the numeric subset; routes everything through the LLM regardless of determinism.
B4 – Fixed hybrid	Rule engine and RAG+LLM present, but routed by a hand-authored fixed rule_type → route table, with no learning or per-query adaptation.	Same reasoning engines as the proposed system; isolates the value of adaptive routing specifically.
Proposed	Rule engine + RAG+LLM + human-review escalation, governed by the adaptive router (Section VIII) with the hard-override grounding guarantee.	Bounded by extraction quality (RQ3) and by benchmark realism (Section XIX).

TABLE VIII
BASELINES COMPARED AGAINST THE PROPOSED SYSTEM (REGULATION COMPREHENSION).

Baseline	Description	Known limitation
C1 – Keyword search	Lexical (BM25-style) search over raw extracted document text, with no clause structuring, no embeddings, and no generation; the user reads the retrieved passage directly.	No answer synthesis or simplification; fails on paraphrased queries with no lexical overlap with the source clause.
C2 – LLM-only (no retrieval)	The question is answered directly by the LLM from its parametric knowledge, with the raw document supplied in-context only up to a fixed context budget, without clause-level retrieval or citation constraints.	No grounding guarantee; expected to be most exposed to hallucination on long documents that exceed the context budget and on multi-clause questions.
C3 – Generic RAG	Standard retrieval-augmented generation [24] over the document with generic chunking (fixed-length windows), without RBI-specific clause segmentation, definition linking, category classification, or version/effective-date metadata (Section V).	Retrieves topically relevant but structurally misaligned chunks (e.g., splitting a clause mid-sentence); no versioning or cross-reference resolution.
Proposed	Layout-aware ingestion, clause segmentation, definition linking, and category classification (Section V) feeding structured retrieval and citation-constrained generation, with version-aware resolution.	Bounded by extraction quality (RQ3) and by the disclosed scale of the comprehension benchmark (Section XIII.E).

B. Metrics for Regulation Comprehension

Metrics for Module 1 are likewise grouped by the sub-benchmark of Section XIII they are computed against, rather than pooled into a single comprehension score.

A. Extraction quality: clause-segmentation boundary accuracy (fraction of predicted clause boundaries matching the manual annotation exactly); definition-linking precision and recall (whether a defined term is correctly linked to its governing definition); obligation-type classification accuracy and category-classification F1 (both against the manual labels of Section XIII.B), reported per requirement category A–G as in Table IV, since we expect segmentation and typing to be markedly easier for quantitative (category A) clauses than for qualitative (category E) or relational (category F) ones.

B. Retrieval and question-answering quality: retrieval is evaluated independently of generation using top- k recall (whether the gold supporting clause appears among the top- k retrieved clauses) and mean reciprocal rank (MRR),

$$\text{MRR} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{\text{rank}_q}, \quad (7)$$

where rank_q is the rank of the first correctly retrieved clause for question q . End-to-end answer quality is then evaluated using the retrieval-augmented-generation decomposition of

RAGAs [25]: *context precision* and *context recall* (retrieval-side), and *faithfulness* (the fraction of claims in the generated answer that are supported by the retrieved clauses) and *answer relevance* (generation-side). Citation accuracy (whether the cited clause identifier genuinely supports the stated answer, evaluated directly against the gold supporting clause rather than only via faithfulness) and hallucination rate (the fraction of answers citing a non-existent or incorrect clause identifier) are reported using the same definitions as the corresponding compliance-side metrics (Section XV.A), following the broader hallucination-evaluation framing of Ji et al. [26], so that grounding quality is comparable, not merely analogous, across the two modules.

C. Simplification quality: SARI [27], which scores a simplification’s inserted, deleted, and kept n -grams against one or more reference simplifications and is, to our knowledge, the most widely used automatic metric for this purpose; the Flesch–Kincaid Grade Level readability score, to check that the simplification is in fact easier to read rather than merely shorter; and a meaning-preservation check comparing the simplification’s stated conditions and exceptions against the original clause’s, since neither SARI nor a readability score on its own would catch a simplification that reads well but silently drops an exception (Section V.H).

D. Cross-module consistency: the fraction of clauses for

TABLE IX
[SYNTHETIC / PLACEHOLDER] RULE-EXTRACTION ACCURACY BY
REQUIREMENT CATEGORY.

Category	Precision	Recall	F1	Executability
A – Quantitative	0.91*	0.88*	0.89*	0.95*
B – Boolean/proc.	0.87*	0.84*	0.85*	0.92*
C – Conditional	0.79*	0.73*	0.76*	0.81*
D – Temporal	0.85*	0.80*	0.82*	0.88*
E – Qualitative	0.68*	0.71*	0.69*	n/a
F – Relational	0.74*	0.66*	0.70*	0.77*
Overall (macro-avg)	0.81*	0.77*	0.79*	0.87*

which the category label and version metadata used by the question-answering interface (this section) agree exactly with the category label and version metadata used by the compliance decision engine (Section XV.A) for the same underlying rule_id, which should be 1.0 by architectural construction (Section X) since both read the same stored object; a measured value below 1.0 would indicate a synchronization defect between the two modules’ read paths rather than a modeling error, and is reported specifically to make that distinction checkable.

XVI. RESULTS

Placeholder notice. Every value reported in this section is a synthetic, illustrative placeholder inserted only to demonstrate the intended table and figure structure and reporting format, for both modules. These are *not* experimental findings. Every value must be replaced by running the pipelines described in Sections V–XIII against the actual extracted RBI rule corpus and the two benchmarks constructed in Sections XII and XIII, and this section must be regenerated before any submission, sharing, or citation of this manuscript.

A. Results for Adaptive Compliance Checking (Module 2)

1) *Rule-extraction accuracy (RQ3)* – [TO BE REPLACED WITH EXPERIMENTAL RESULT]: Table IX shows the intended reporting format for per-category extraction accuracy. Values marked with an asterisk are illustrative placeholders, not measured results.

The pattern of lower placeholder scores for categories C and F is consistent with the expectation, drawn from the automated compliance checking literature (Section III), that conditional and relational clauses are harder to extract reliably than single-threshold quantitative clauses; whether this pattern replicates in real results is itself a reportable finding once the pipeline is run.

2) *Compliance accuracy and grounding by case type (RQ1, RQ4)* – [TO BE REPLACED WITH EXPERIMENTAL RESULT]: Table X shows the intended reporting format. The qualitative pattern the architecture predicts – a larger proposed-system advantage concentrated in missing-data and adversarial cases, and a smaller advantage on straightforward

TABLE X
[SYNTHETIC / PLACEHOLDER] ACCURACY AND HALLUCINATION
RATE BY CASE TYPE.

Case type	B2 Acc.	B3 Acc.	Proposed Acc.	Proposed Halluc.
Compliant/Non-compl.	0.90*	0.93*	0.98*	0.01*
Boundary	0.71*	0.78*	0.96*	0.02*
Near-boundary	0.66*	0.74*	0.93*	0.03*
Ambiguous	0.58*	0.69*	0.81*	0.06*
Missing-data	0.41*	0.55*	0.92*	0.04*
Adversarial	0.35*	0.49*	0.88*	0.05*
Contradictory	0.47*	0.58*	0.84*	0.07*
Cross-version	0.62*	0.70*	0.90*	0.03*

TABLE XI
[SYNTHETIC / PLACEHOLDER] ROUTING COMPARISON ACROSS
R1–R4.

Router	Acc.	LLM calls avoided	Latency (ms)	Cost/case
R1 Static rule-first	0.86*	41%*	310*	\$0.004*
R2 Confidence-threshold	0.89*	52%*	260*	\$0.006*
R3 Fixed hybrid	0.90*	58%*	240*	\$0.007*
R4 Proposed adaptive	0.94*	67%*	205*	\$0.008*

compliant/non-compliant cases – should be checked for once real data is available, not assumed; a uniform win across all case types with no differentiation would itself be a warning sign that the benchmark is not discriminating between systems.

3) *Routing cost-efficiency (RQ2)* – [TO BE REPLACED WITH EXPERIMENTAL RESULT]: Table XI shows the intended reporting format. The headline claim intended to be tested is that R4 Pareto-dominates R1–R3 – higher accuracy at lower latency, with a modest cost increase over R1–R3 driven by more selective, higher-value LLM invocation – rather than winning on accuracy alone. Once real results are available, the full frontier should be reported, for example as a scatter plot of accuracy against cost with each router represented as a point, rather than only the tabular summary below.

B. Results for Regulation Comprehension (Module 1)

Because the paper’s evaluation treats both modules as first-class, equally weighted deliverables (Section V), Module 1 results are reported in the same placeholder format and with the same caution as Module 2: every figure below is synthetic and must be replaced once the comprehension pipeline (Section V) is run against the benchmark of Section XIII.

1) *Extraction quality by requirement category (RQ6)* – [TO BE REPLACED WITH EXPERIMENTAL RESULT]: Table XII shows the intended reporting format for clause-segmentation and category-classification accuracy, broken down by requirement category as in Table IX for consistency across the two modules.

Segmentation boundary accuracy is placeholder-projected as consistently higher than obligation-type or category accuracy across all requirement categories, reflecting that segmentation is a largely structural task (Section V.D) while obligation-type

TABLE XII
[SYNTHETIC / PLACEHOLDER] COMPREHENSION EXTRACTION
QUALITY BY REQUIREMENT CATEGORY.

Category	Segmentation boundary acc.	Obligation-type accuracy	Category F1
A – Quantitative	0.97*	0.93*	0.90*
B – Boolean/proc.	0.96*	0.90*	0.88*
C – Conditional	0.92*	0.82*	0.83*
D – Temporal	0.94*	0.87*	0.86*
E – Qualitative	0.91*	0.75*	0.78*
F – Relational	0.89*	0.71*	0.74*
Overall (macro-avg)	0.93*	0.83*	0.83*

and category labeling require semantic judgment; whether the gap is as pronounced as shown here is itself a question the real experiment must answer.

2) *Retrieval and question-answering quality (RQ6)* – [TO BE REPLACED WITH EXPERIMENTAL RESULT]: Table XIII shows the intended reporting format across the three comprehension baselines of Table VIII and the proposed system, using the retrieval and RAGAs-style [25] metrics defined in Section XV-B.B.

The placeholder pattern shown – a larger proposed-system advantage in context precision and faithfulness than in answer relevance – is intended to reflect that RBI-specific clause structuring (Section V) should primarily improve *which* text is retrieved and *how well-grounded* the answer is, rather than the fluency of the answer itself, which even the LLM-only baseline can achieve; this is a hypothesis the real experiment must test, not an assumed result.

3) *Simplification quality (RQ7)* – [TO BE REPLACED WITH EXPERIMENTAL RESULT]: Table XIV shows the intended reporting format for the simplification sub-benchmark (Section XIII.D), using SARI [27] and Flesch–Kincaid Grade Level alongside the meaning-preservation check of Section XV-B.C.

The placeholder meaning-preservation scores are projected lowest for category E (qualitative) clauses, consistent with the intuition that a qualitative standard’s nuance is the most fragile under simplification; this is precisely the reason meaning preservation is measured explicitly (Section XV-B.C) rather than relying on SARI or FKGL alone, both of which would report a favorable score for a simplification that dropped a qualitative nuance as noise.

4) *Cross-module consistency (RQ6)* – [TO BE REPLACED WITH EXPERIMENTAL RESULT]: As defined in Section XV-B.D, the cross-module consistency check is expected to equal 1.0 by architectural construction (Section X); its placeholder value here is [1.0*, i.e., no known reason for it to differ], and it is reported specifically as a sanity check to be run once both pipelines are implemented, not as a metric expected to show interesting variation.

XVII. ABLATION STUDY

A. Ablations for Adaptive Compliance Checking

Table XV shows the intended reporting format for the planned ablation study; no findings are reported yet.

The four planned ablations are intended to isolate, respectively: the marginal contribution of adaptive routing over a fixed hybrid decision table (reduces to B4); the contribution of the deterministic rule engine over retrieval-augmented reasoning alone (reduces to B3); the contribution of evidence grounding over ungrounded LLM reasoning (reduces to B2); and the contribution of the deterministic validation step in rule extraction (Section V), disabled in the fourth configuration. Ablations isolating a “no graph module” condition and a “no confidence mechanism” condition are omitted, since no graph module is implemented (Section IV.A) and the confidence-threshold comparison is already captured by baseline router R2 (Section VIII.D) rather than warranting a separate ablation. No numerical finding is claimed in this section; the table above states the intended reporting format only.

B. Ablations for Regulation Comprehension

A parallel set of ablations isolates the contribution of the comprehension-specific design choices of Section V, shown in Table XVI.

These ablations are intended to isolate, respectively: the contribution of layout-aware ingestion (Section V.B) over flat-text extraction; the contribution of RBI-specific clause segmentation over generic fixed-length chunking (reduces retrieval to baseline C3, Table VIII); the contribution of definition linking (Section V.E) specifically to simplification quality, since an unlinked defined term is a common source of an under-specified simplification; and the contribution of the citation constraint on generation (Section V.N) to faithfulness, with retrieval metrics reported as not applicable in that configuration since removing the citation constraint does not by itself change what is retrieved. As with Table XV, no numerical finding is claimed; the table states the intended reporting format only.

XVIII. NOVELTY AND CONTRIBUTION ANALYSIS

Each candidate contribution is assessed against the related work of Section III individually, rather than asserting novelty uniformly across the system, consistent with the principle that implementation technology – a specific database or a specific rule-engine library – is not itself a research contribution. Table XVII summarizes the assessment.

The paper’s contribution claim rests principally on Section VIII (the router, to be evaluated as described in Section XVI) and secondarily on Sections V and XII (extraction and benchmark, to be evaluated as described in Section XVI), and not on the system’s use of any particular database or rule-engine library. What is explicitly not claimed as novel includes generic retrieval-augmented generation, large language model use for compliance reasoning in general, deterministic rule engines, rule extraction as a technique, a hybrid rule-plus-LLM architecture in general, and generic LLM-to-LLM routing.

TABLE XIII
[SYNTHETIC / PLACEHOLDER] RETRIEVAL AND QUESTION-ANSWERING QUALITY ACROSS COMPREHENSION BASELINES.

System	Top-5 recall	MRR	Context precision	Context recall	Faithfulness	Answer relevance
C1 – Keyword search	0.58*	0.41*	0.52*	0.55*	n/a	n/a
C2 – LLM-only (no retrieval)	n/a	n/a	n/a	n/a	0.61*	0.74*
C3 – Generic RAG	0.79*	0.68*	0.71*	0.75*	0.82*	0.85*
Proposed (RBI-structured RAG)	0.94*	0.87*	0.90*	0.92*	0.95*	0.93*

TABLE XIV
[SYNTHETIC / PLACEHOLDER] SIMPLIFICATION QUALITY BY REQUIREMENT CATEGORY.

Category	SARI	FKGL (source → simplified)	Meaning preserved
A – Quantitative	44.2*	15.1 → 9.8*	0.97*
B – Boolean/proc.	42.7*	14.6 → 9.4*	0.96*
C – Conditional	39.5*	16.3 → 11.2*	0.90*
D – Temporal	41.8*	14.9 → 9.9*	0.94*
E – Qualitative	35.1*	17.5 → 12.8*	0.85*
Overall (macro-avg)	40.7*	15.7 → 10.6*	0.92*

TABLE XV
[SYNTHETIC / PLACEHOLDER] COMPLIANCE-CHECKING ABLATION CONFIGURATIONS AND INTENDED REPORTING FORMAT.

Configuration	Acc.	Halluc. rate	Latency (ms)
Full proposed system	0.94*	0.03*	205*
– adaptive routing (→ B4)	0.90*	0.04*	240*
– rule engine (→ B3)	0.79*	0.09*	290*
– evidence grounding (→ B2)	0.65*	0.18*	180*
– extraction validation	0.87*	0.06*	210*

XIX. LIMITATIONS AND THREATS TO VALIDITY

A. Threats to validity

Rule-extraction verification is intended to cover a bounded, disclosed sample (Section XII), not the full RBI regulatory corpus; extraction accuracy figures, once measured, should not be read as applying uniformly to all RBI regulation. Large language model choice affects reproducibility: extraction and retrieval-augmented-reasoning results are expected to shift if the underlying model is updated or replaced; model identity and version are to be pinned and reported (Section XII) specifically so this dependency is auditable rather than hidden. The router’s learned component is trained and tuned on the same benchmark family used for evaluation, via a held-out validation split; results should be read as within-benchmark generalization, not as a claim of generalization to regulatory domains outside RBI banking regulation.

B. No real operational bank data

All quantitative results reported once the pipeline is evaluated will be obtained on the synthetic benchmark of Section XII, not on real bank operational data, because such data is confidential, subject to regulatory restriction, and not available

TABLE XVI
[SYNTHETIC / PLACEHOLDER] REGULATION-COMPREHENSION ABLATION CONFIGURATIONS AND INTENDED REPORTING FORMAT.

Configuration	Top-5 recall	Faithful- ness	SARI
Full proposed system	0.94*	0.95*	40.7*
– layout-aware ingestion (→ flat text)	0.87*	0.91*	39.2*
– clause segmentation (→ fixed-length chunks, C3)	0.79*	0.82*	34.5*
– definition linking	0.91*	0.88*	33.1*
– citation constraint (→ C2)	n/a	0.61*	40.1*

for academic research. Every result should be interpreted as a controlled evaluation of the system’s reasoning behavior against known, rule-grounded ground truth, not as a claim about accuracy on live bank data. This limitation is structural to the problem domain, not specific to this system, and is shared by essentially any academic work in this space that cannot obtain a data-sharing agreement with a regulated financial institution.

C. Bounded rule-extraction verification and scope exclusions

Connected-counterparty graph reasoning (category G of Section VI) is not implemented or evaluated. Optical character recognition and scanned-document ingestion are outside the evaluated scope (Section IV.A). Both are discussed as future work in Section XX rather than presented as implemented capabilities.

D. Regulatory drift and deployment considerations

RBI regulation is amended and superseded on an ongoing basis. The version, effective_from, and effective_to fields, together with the cross-version benchmark cases (Section XII), test whether the system’s versioning logic behaves correctly on known amendments, but a deployed system would additionally require a live re-ingestion pipeline to detect and process new amendments as they are published – out of scope for this paper’s evaluation, which operates on a fixed regulatory snapshot.

E. Security and privacy considerations

A system operating on bank-submitted documents in a deployed setting would need to address prompt-injection risk from maliciously or adversarially crafted submitted documents, and would need controls around data leakage of confidential bank information through logging, retrieval indices, or LLM API calls. These are deployment-time engineering

TABLE XVII
NOVELTY AND CONTRIBUTION ANALYSIS.

Claim	Verdict	Basis
Hybrid rule-engine + RAG + LLM architecture for compliance	Not novel in isolation	Established pattern in recent work: knowledge-graph-augmented RAG compliance [5], neuro-symbolic compliance reasoning [6], compliance-as-code [4], and code-generation-based compliance checking [8].
Automated extraction of rules from regulatory text	Not novel as a technique; novel as an evaluated RBI-specific instance	A well-established lineage in automated compliance checking for construction codes [1]–[3]; no equivalent RBI-evaluated system located (Section III.D).
Adaptive reasoning-modality router with a formal cost objective, evaluated against routing baselines	Primary defensible contribution	Structurally distinct from the populated LLM-model-routing literature [10]–[16], which routes between LLMs rather than between reasoning modalities; not found applied to compliance-modality selection.
A single regulatory knowledge base reused by both a human-facing regulation comprehension interface and a machine-facing compliance decision engine	System-level contribution supporting the primary claim	Not itself claimed as a new technique; supports deployability of the router (Section X).
Rule-grounded synthetic RBI compliance benchmark with graded case-difficulty categories, intended for public release	Defensible secondary contribution	No public RBI-specific benchmark located; construction methodology aligned with recent calls for richer benchmark case coverage [17]–[19].
Connected-counterparty graph reasoning for exposure aggregation	Not claimed as a validated contribution	Not implemented; would require synthetic ownership data generated by the authors, judged insufficiently falsifiable to report as a result (Sections IV.A, XX).
RBI-specific regulation comprehension (structured extraction, versioning, citation-grounded QA over Master Directions/circulars)	Not novel as a technique; novel as an RBI-evaluated, jointly-integrated instance	Retrieval-augmented generation [24], layout-aware document understanding [23], and RAG evaluation methodology [25] are all established; no equivalent system evaluated jointly on RBI comprehension and RBI compliance checking located (Section III.D).

concerns rather than properties evaluated in this research paper, and are noted here for completeness rather than addressed empirically.

XX. FUTURE WORK

Realistic extensions beyond the scope evaluated in this paper include: obtaining real bank operational data through secure institutional partnerships, subject to appropriate confidentiality and regulatory approval, to validate results beyond the synthetic benchmark of Section XII; a live RBI regulatory-update pipeline that detects and re-ingests amendments as they are published, extending the offline processing of Section XI; improved handling of scanned and image-based regulatory or evidence documents through optical character recognition, currently excluded (Section IV.A); multilingual regulation comprehension, extending Section V to non-English regulatory or evidentiary text; connected-counterparty graph reasoning for exposure aggregation (category G, Section VI), sketched conceptually below; a richer regulatory knowledge graph capturing cross-document relationships beyond the cross-reference handling already described in Section V; continuous learning for the router’s cost-minimizing policy as new labeled routing outcomes accumulate in deployment; stronger human-in-the-loop workflows for the REQUIRES_HUMAN_REVIEW path, including reviewer feedback that updates `historical_router_accuracy` (Table V); secure deployment architecture for confidential bank data, addressing the concerns noted in Section XIX.E; and broader coverage of Indian financial regulation beyond RBI banking regulation, such as SEBI or IRDAI requirements, where the underlying architecture may transfer but has not been evaluated.

A. Connected-counterparty graph reasoning (theoretical sketch)

RBI’s large-exposure framework requires aggregating exposure across connected counterparties – parent/subsidiary, common ownership, guarantee, and economic-dependency relationships. This subsystem is not implemented or evaluated in this work. As a theoretical sketch only: entities and relationships would be represented as a property graph, with nodes as legal entities and edges as ownership, control, or guarantee relationships weighted by ownership percentage where available; connected-component or community-detection algorithms would identify counterparty groups; exposure-aggregation rules (categories F and G of Section VI) would then apply the existing quantitative rule logic to the aggregated group rather than the individual entity. This is not implemented because a meaningful evaluation would require multi-hop ownership data not publicly available, and evaluating the subsystem against ownership data generated by the authors themselves would not constitute independent validation. This is flagged explicitly as future work requiring either a data-sharing arrangement with a financial institution or a carefully justified, separately validated synthetic-graph methodology beyond the scope of this paper.

XXI. CONCLUSION

This paper presented an integrated regulatory intelligence system for RBI banking regulation, consisting of a Regulation Comprehension module that converts RBI circulars and Master Directions into structured, searchable, citation-grounded regulatory knowledge, and an Adaptive Compliance Checking

module that reuses this same knowledge to evaluate bank-submitted evidence from RBI’s supervisory perspective. The system’s primary contribution is a reasoning-modality router that selects between a deterministic rule engine, a retrieval-grounded large language model, and human escalation under an explicit cost–accuracy–latency objective, a problem we distinguish from the well-studied LLM-to-LLM model-routing literature. This is supported by a schema-constrained rule-extraction pipeline for RBI regulatory text, a systematically constructed rule-grounded synthetic benchmark covering nine case-difficulty categories, and an evidence-grounded explanation architecture that traces every decision and every comprehension answer to a specific rule, clause, and evidentiary or source input. We evaluated the system’s design against four baselines and four planned ablations; the results reported in Section XVI are, at the time of writing, synthetic placeholders and must be replaced with real experimental results from the pipeline described in Sections V–XII before this manuscript can be considered complete. We have been explicit throughout about what is and is not novel (Section XVIII) and about what is not implemented or evaluated, including connected-counterparty graph reasoning (Section XX) and optical character recognition (Section IV.A), on the view that a narrower, honestly evaluated contribution is more defensible than a broader, under-evaluated one.

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